

Behnam Yousefimehr

PH.D. IN COMPUTER SCIENCE — ARTIFICIAL INTELLIGENCE · ADJUNCT LECTURER AT AMIRKABIR UNIVERSITY OF TECHNOLOGY AND
MOHAGHEGH ARDABILI UNIVERSITY

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Summary

Behnam Yousefimehr holds a Ph.D. in Computer Science with a specialization in Artificial Intelligence and Soft Computing from Amirkabir University of Technology (Tehran Polytechnic). His research interests include artificial intelligence, machine learning, data mining, and complex data analysis, with a particular focus on anomaly detection and the development of intelligent models for solving real-world problems. He has published numerous papers in reputable international journals and conferences and has experience in both academic and applied research projects.

In addition to his academic activities, he works as an Artificial Intelligence Specialist at Tafahom Artificial Intelligence Technology Development Company. In this role, he is involved in the design, development, and implementation of intelligent systems and machine learning-based solutions for industrial projects. He has also collaborated with universities and research centers and is active in applying AI and data analytics technologies to real-world industrial applications.

Education

Mohaghegh Ardabili University

B.SC. IN COMPUTER SCIENCE

Ardabil, Iran

Oct 2016 – Feb 2020

Amirkabir University of Technology (Tehran Polytechnic)

M.SC. IN COMPUTER SCIENCE (DATA MINING)

Tehran, Iran

Sep 2020 – Sep 2022

Amirkabir University of Technology (Tehran Polytechnic)

PH.D. IN COMPUTER SCIENCE (ARTIFICIAL INTELLIGENCE & SOFT COMPUTING)

Tehran, Iran

Sep 2022 – Apr 2026

Experience

Tafahom Artificial Intelligence Technology Development Company

ARTIFICIAL INTELLIGENCE SPECIALIST

Tehran, Iran

Apr 2026 – Present

- Served as Project Manager for the Pasargad Upstream Intelligent Assistant project, overseeing project planning, technical coordination, and AI solution development.
- Developed an oil production forecasting solution using the Chronos-2 foundation model for time-series forecasting and production data analysis.
- Supervised outsourced AI projects, monitoring technical progress, deliverables, and alignment with project requirements and organizational objectives.
- Provided AI consulting on model selection, solution architecture, feasibility assessment, and the application of machine learning and foundation models to industrial use cases.

Research Center for Optimization, Networking, and Smart City

SENIOR RESEARCHER

Tehran, Iran

Oct 2020 – Present

- Served as Internal Project Manager for the “Optimizing Urban Bridge Inspection Processes in Tehran using AI” project, conducted in collaboration with the Tehran Urban Planning and Research Center (TUPRC).
- Coordinated project execution, technical activities, research tasks, and communication among internal project stakeholders.

Amirkabir University of Technology (Tehran Polytechnic)

ADJUNCT LECTURER

Tehran, Iran

Oct 2022 – Present

- Advanced Programming
- Artificial Intelligence
- Data Mining

Mohaghegh Ardabili University

ADJUNCT LECTURER

Ardabil, Iran

Oct 2025 – Present

- Advisor to graduate theses

Iranian Research Institute for Information Science & Technology (IRANDOC)

COURSE INSTRUCTOR

Tehran, Iran

2023

- Delivered specialized training courses in Artificial Intelligence

Honors & Awards

2026	Overall 2nd Place — Artificial Intelligence Competitions , AI Evaluation Reference Laboratory, ICT Research Institute (ITRC)	Iran
2026	1st Place — Electrofans Bearing Fault Detection Using Vibration Data , AI Evaluation Reference Laboratory, ICT Research Institute (ITRC)	Iran
2026	3rd Place — Agricultural Land Cadastral Detection Using Satellite Imagery and AI , AI Evaluation Reference Laboratory, ICT Research Institute (ITRC)	Iran
2025	Awardee of Shahid Vezvaei, Shahid Sayyad Shirazi, Shahid Ahmadi Roshan, and Pazhoheshyar programs by the National Elites Foundation (NEF) , National Elites Foundation (NEF)	Iran
2025	Selected as Outstanding Researcher and Student , National Elites Foundation (NEF)	Iran
2024	2nd Place — Shatel Customer Loyalty Prediction Challenge , Amirkabir University AI Competitions (AAIC)	Tehran, Iran
2024	Diploma of Honor , Student Research Symposium	Iran
2023	1st Place — Reflux Detection Using pH-Impedance Data , Isfahan NEF (IAI) AI Competition	Isfahan, Iran
2023	Overall 2nd Place , Isfahan NEF (IAI) AI Competition	Isfahan, Iran
2022	Direct Admission (No Entrance Exam) to Ph.D. in Computer Science , Amirkabir University of Technology (Tehran Polytechnic)	Tehran, Iran
2022	Ranked 1st among M.Sc. Computer Science Students , Amirkabir University of Technology (Tehran Polytechnic)	Tehran, Iran
2020	Direct Admission (No Entrance Exam) to M.Sc. in Computer Science , Amirkabir University of Technology (Tehran Polytechnic)	Tehran, Iran
2019	Ranked 1st among B.Sc. Computer Science Students , Mohaghegh Ardabili University	Ardabil, Iran
2018	Top Rank — 5th Provincial Programming Contest , Mohaghegh Ardabili University	Ardabil, Iran

Selected Projects

Amirkabir University of Technology (Tehran Polytechnic) FRAUD DETECTION VIA HYBRID RESAMPLING AND ENSEMBLE LEARNING	Tehran, Iran Oct 2021 – Sep 2022
Amirkabir University of Technology (Tehran Polytechnic) DISTRIBUTION-PRESERVING ANOMALY DETECTION USING HYBRID ARCHITECTURES AND KNOWLEDGE DISTILLATION	Tehran, Iran Oct 2022 – Apr 2026
Tehran Urban Planning and Research Center (TUPRC) OPTIMIZING URBAN BRIDGE INSPECTION PROCESSES IN TEHRAN USING AI	Tehran, Iran Dec 2024 – Present

Skills

Programming	Python, C, C++, MATLAB
Technical	LaTeX, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, LLM, RAG, Agentic AI
Languages	Persian, English, Turkish

Publications

[1] B. Yousefimehr, M. Ghathe, and R. Razavi-Far, “Multi-teacher knowledge distillation framework for lightweight anomaly detection,” *Neural Networks*, p. 108267, 2025.

[2] B. Yousefimehr, M. Ghathe, and R. Razavi-Far, “Multi-stage self-distillation for class-imbalanced anomaly detection,” *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2026.

[3] B. Yousefimehr and M. Ghathe, “A distribution-preserving method for resampling combined with lightgbm-lstm for sequence-wise fraud detection in credit card transactions,” *Expert Systems with Applications*, vol. 262, p. 125661, 2025.

- [4] B. Yousefimehr and M. Ghathe, "A systematic survey and empirical comparison of hybrid methods for imbalanced fraud detection: Combining resampling and machine learning," *AUT Journal of Mathematics and Computing*, 2025.
- [5] B. Yousefimehr, M. Ghathe, S. Ghaffari, A. Arasteh, P. Ahmadi, A. Ghane, and S. Esnaasharieh, "A systematic review for 2019–2025 on deep learning models in the film production industry," *Entertainment Computing*, p. 101076, 2025.
- [6] S. Saronian, B. Yousefimehr, M. Ghathe, and M. M. Bejani, "An ensemble of deep clustering models with autoencoders to mine travel patterns from smart card data," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [7] Y. Behroo, B. Yousefimehr, and R. Parvaz, "A new 2d chaotic system and application in hash function by cipher block chaining," *International Journal of Information Security Science*, vol. 15, no. 2, p. 1–28, 2026.
- [8] B. Yousefimehr, M. Ghathe, S. Moradi, Y. Tafakor, and S. Tavakoli, "Detecting car insurance fraud using improved clustering with genetic algorithm," *Iranian Journal of Insurance Research*, vol. 14, no. 2, pp. 109–118, 2025.
- [9] B. Yousefimehr and M. Ghathe, "Trustworthy ai for unbalanced industrial data classification," in *2026 International Biennial Conference on Advances in Artificial Intelligence and Data Science (IBAAIDS)*, pp. 1–5, IEEE, 2026.
- [10] A. R. Kenari, A. Montazerolghaem, Z. Zojaji, M. Ghathe, B. Yousefimehr, A. Rahmani, M. Kalani, F. Kiyandpour, M. Kiani-Abari, M. Y. Fakhar, *et al.*, "Isfahan artificial intelligence event 2023: Reflux detection competition," *Journal of Medical Signals & Sensors*, vol. 15, no. 2, p. 6, 2025.
- [11] S. Yousefi, S. Najjar-Ghabel, B. Yousefimehr, M. Kozhanov, N. Nabipour, and A. Amirov, "Internet of educational things and machine learning," *Chart*, p. 11, 2026.
- [12] B. Yousefimehr, M. Ghathe, and A. Heydari, "Improving adhd detection with cost-sensitive lightgbm," in *2024 14th International Conference on Computer and Knowledge Engineering (ICCKE)*, pp. 109–113, IEEE, 2024.
- [13] Z. Rafei, S. F. Hosseini, B. Yousefimehr, and M. Ghathe, "Drug-disease association analysis via machine learning on extracted features by matrix decomposition," *Mathematics Interdisciplinary Research*, vol. 10, no. 3, pp. 295–313, 2025.
- [14] S. Ghaffari, B. Yousefimehr, and M. Ghathe, "Generative-ai in e-commerce: Use-cases and implementations," in *2024 20th CSI International Symposium on Artificial Intelligence and Signal Processing (AISIP)*, pp. 1–5, IEEE, 2024.
- [15] A. Daneshpour, B. Yousefimehr, and M. Ghathe, "Balanced learning with optimized extra trees classifier for reliable lithology identification in imbalanced well log data," in *2025 15th International Conference on Computer and Knowledge Engineering (ICCKE)*, pp. 1–6, 2025.
- [16] M. Moghaddam, A. Daneshpour, B. Yousefimehr, and M. Ghathe, "Causally informed domain-guided machine learning for imbalanced lithofacies classification from well logs," in *2026 International Biennial Conference on Advances in Artificial Intelligence and Data Science (IBAAIDS)*, pp. 1–6, IEEE, 2026.
- [17] A. Azizian, B. Yousefimehr, and M. Ghathe, "Enhanced multi-modal gas leakage detection with nsmote: A novel over-sampling approach," in *2024 8th international conference on smart cities, internet of things and applications (SCIoT)*, pp. 94–99, IEEE, 2024.
- [18] B. Yousefimehr, M. Ghathe, M. A. Seifi, J. Fazli, S. Tavakoli, Z. Rafei, S. Ghaffari, A. Nikahd, M. R. Gandomani, A. Orouji, *et al.*, "Data balancing strategies: A survey of resampling and augmentation methods," *arXiv preprint arXiv:2505.13518*, 2025.
- [19] Z. Rafei, S. F. Hosseini, B. Yousefimehr, S. Tavakkoli, and M. Ghathe, "Optimizing drug-disease association analysis: A resource-efficient approach using numerical linear algebra and machine learning," in *First International Conference on Machine Learning and Knowledge Discovery (MLKD 2024)*, 2024.
- [20] N. Falah, B. Yousefimehr, and M. Ghathe, "Predicting music track popularity by convolutional neural networks on spotify features and spectrogram of audio waveform," *arXiv preprint arXiv:2505.07280*, 2025.
- [21] S. Torkhesari, B. Yousefimehr, and M. Ghathe, "Car sensors health monitoring by verification based on autoencoder and random forest regression," *arXiv preprint arXiv:2505.00876*, 2025.